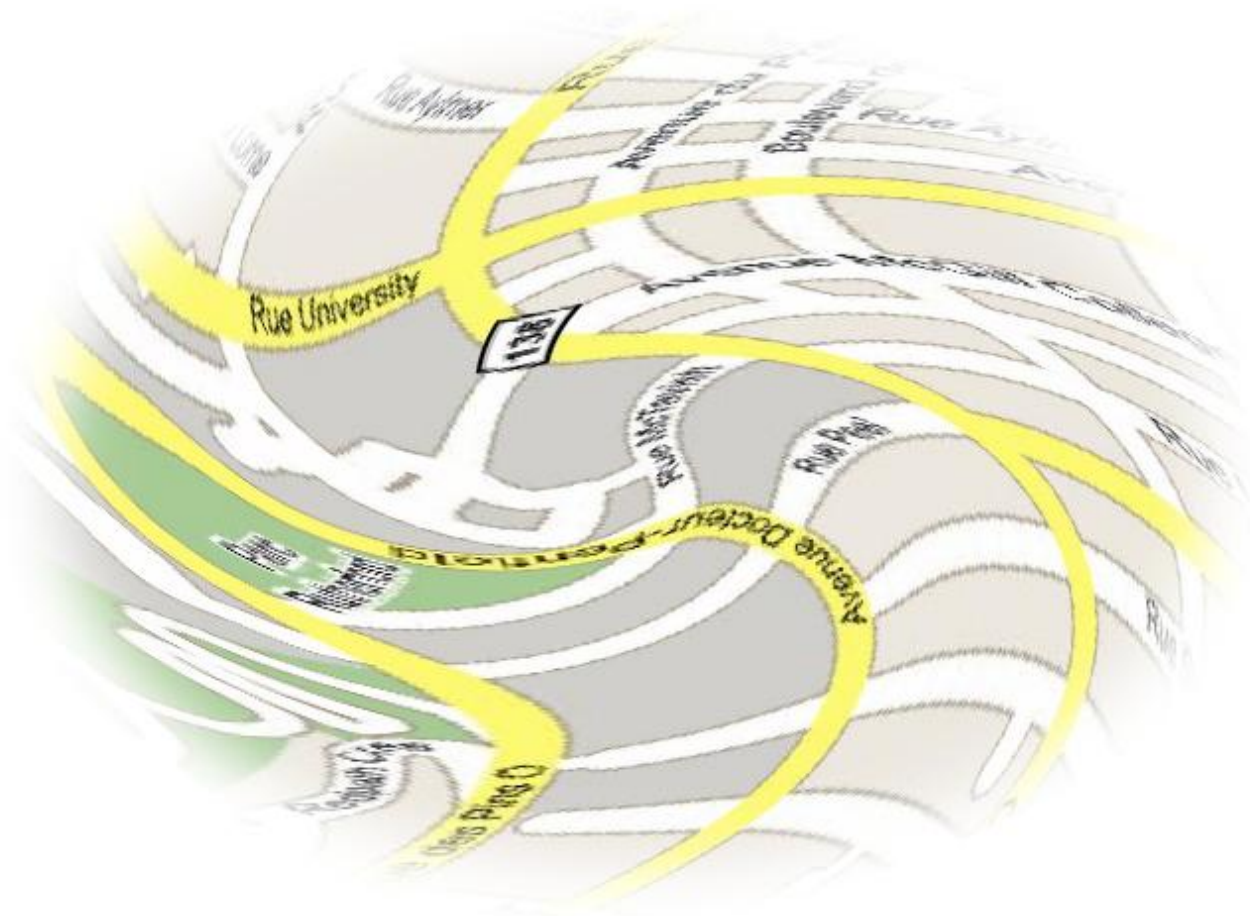


Action Respecting Embedding



April 25, 2005

Michael Bowling
University of Alberta

Acknowledgments

- Joint work with...

Ali Ghodsi and Dana Wilkinson.

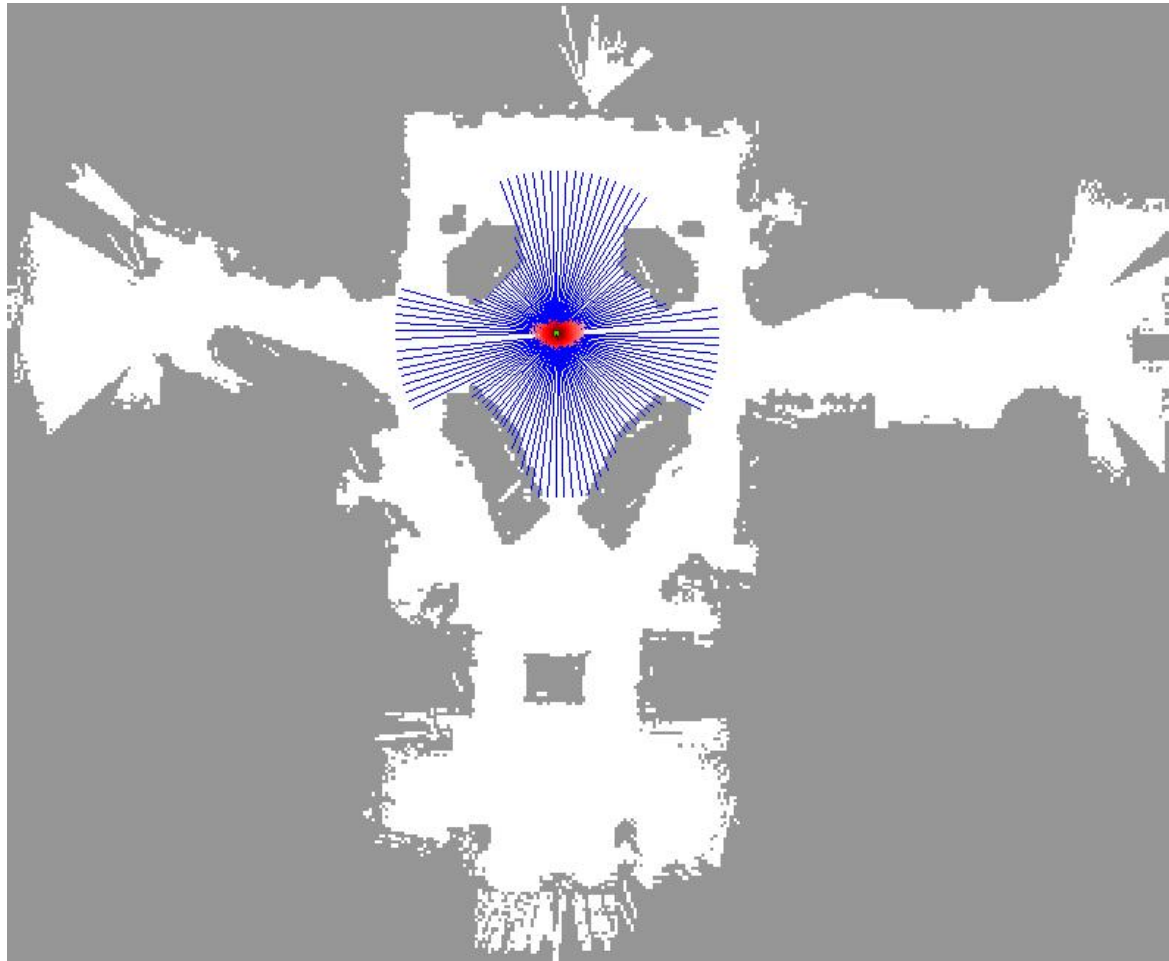
- Localization work with...

Adam Milstein and Wesley Loh.

- Discussions and insight...

Finnegan Southey, Dale Schuurmans, Tao Wang,
Dan Lizotte, Michael Littman, and Pascal Poupart.

What is a Map?



Robot Maps

- Maps are Models.
 - Motion: $P(x_{t+1}|x_t, u_t)$.
 - Sensor: $P(z_t|x_t)$.
- Robot Pose: $x_t = (x_t, y_t, \theta_t)$.
 - **Objective** representation.
 - Models relate actions (u_t) and observations (z_t) to this frame of reference.
- Can a map be learned with only **subjective** experience?

x_t : pose

u_t : action or control

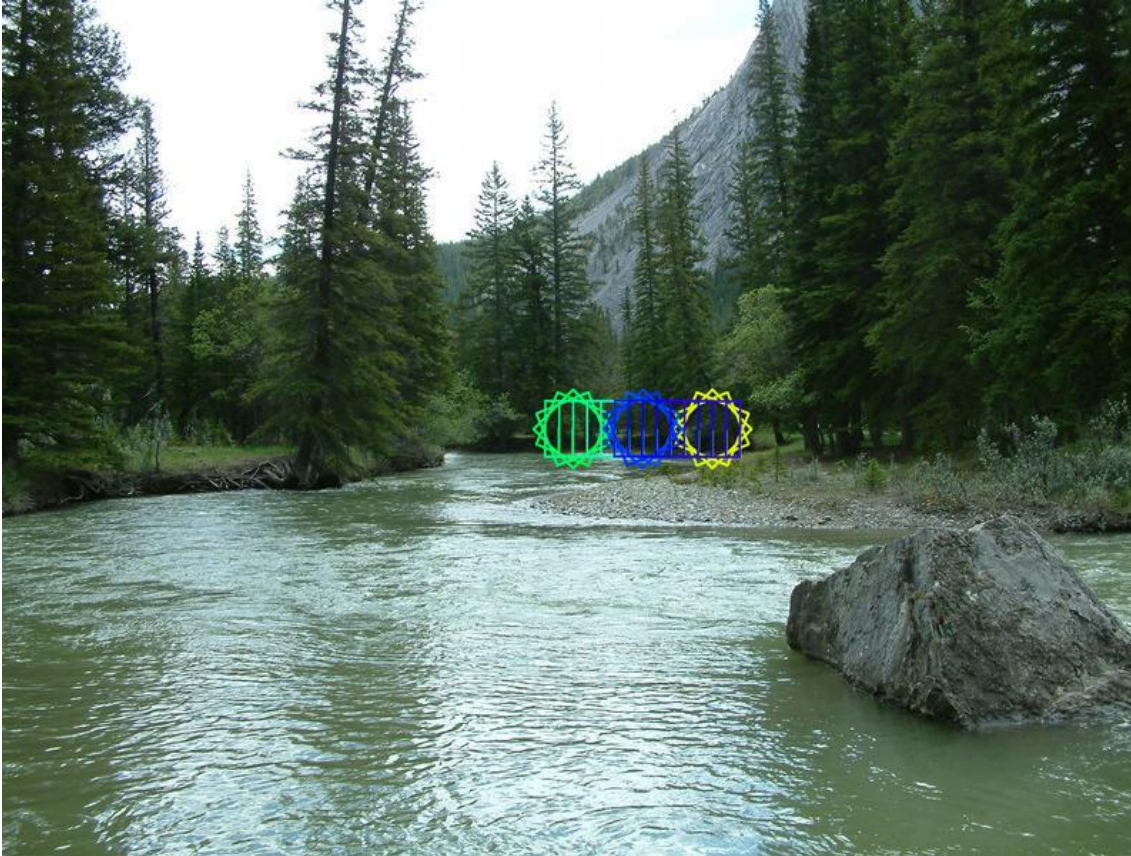
z_t : observation

$z_1, u_1, z_2, u_2, \dots, u_{T-1}, z_T$

Not an objective map.

ImageBot

- A “Robot” moving around on a large image.

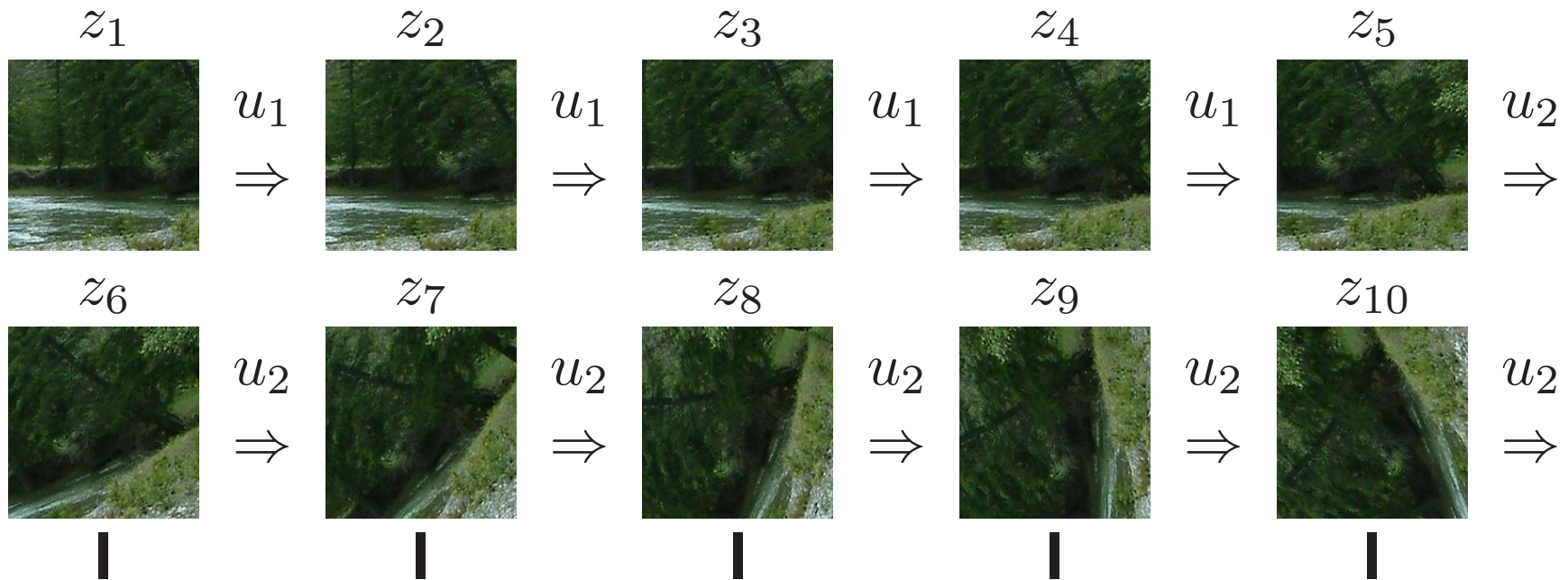


- Forward
- Backward
- Left
- Right
- Turn-CW
- Turn-CCW
- Zoom-In
- Zoom-Out

- Example: $F \times 5, CW \times 8, F \times 5, CW \times 16, F \times 5, CW \times 8$

ImageBot

- Construct a map from a stream of input.



- Actions are labels with no semantics.
- No image features, just high-dimensional vectors.
- Can a map be learned with only **subjective** experience?

Overview

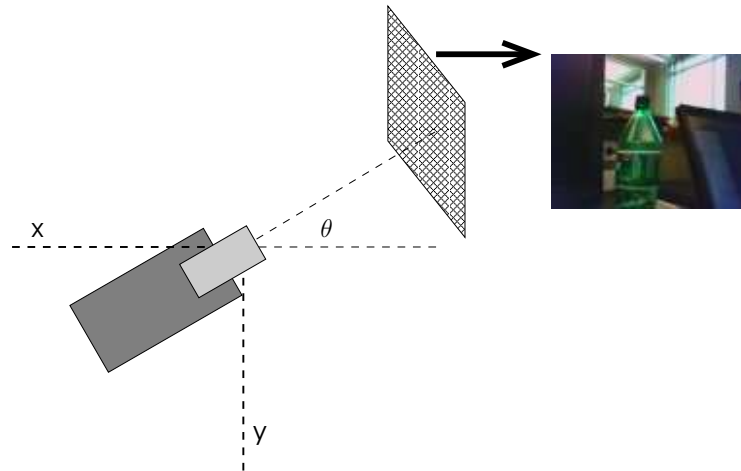
- What is a Map?
- Subjective Mapping
- Action Respecting Embedding (ARE)
- Results
- Future Work

Subjective Maps

- What is a **subjective map**?
 - Allows you to do “map things”, e.g., localize and plan.
 - No models.
 - Representation (i.e., pose) can be anything.
- Subjective mapping becomes choosing a representation.
 - What is a good representation?
 - How do we extract it from experience?
 - How do we answer our map questions with it?

Subjective Representation

- (x, y, θ) is often a good representation. Why?
 - Sufficient representation for generating observations.



- Low dimensional (despite high dimensional observations).
- Actions are simple transformations.

$$x_{t+1} = (x_t + F \cos \theta_t, \quad y_t + F \sin \theta_t, \quad \theta_t + R)$$

Subjective Representation

- How do we extract a representation like (x, y, θ) ?
 - Low dimensional description of observations?

Dimensionality Reduction

- Respects actions as simple transformations?

Semidefinite Embedding (SDE)

(Weinberger & Saul, 2004)

- Goal: Learn the kernel matrix, K , from the data.
- Optimization Problem:

Maximize: $\text{Tr}(K)$

Subject to: $K \succeq 0$

$$\sum_{ij} K_{ij} = 0$$

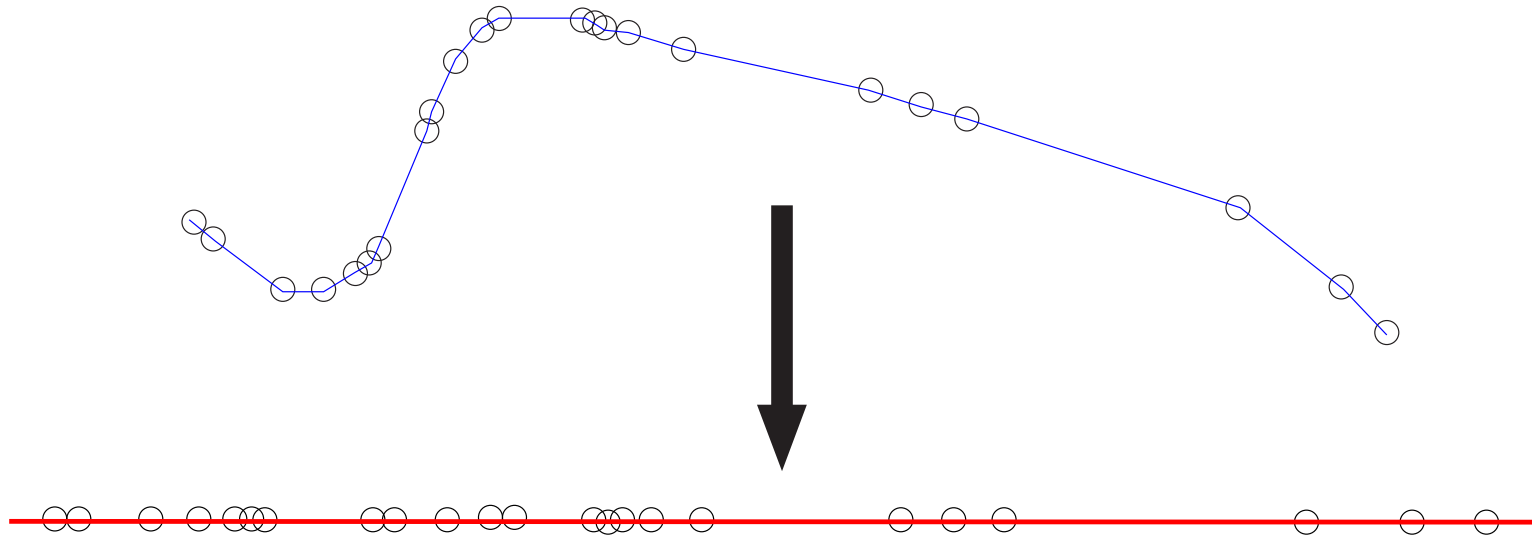
$$\forall i, j \quad \eta_{ij} > 0 \vee [\eta^T \eta]_{ij} > 0 \Rightarrow$$

$$K_{ii} - 2K_{ij} + K_{jj} = \|x_i - x_j\|^2$$

where η comes from k -nearest neighbors.

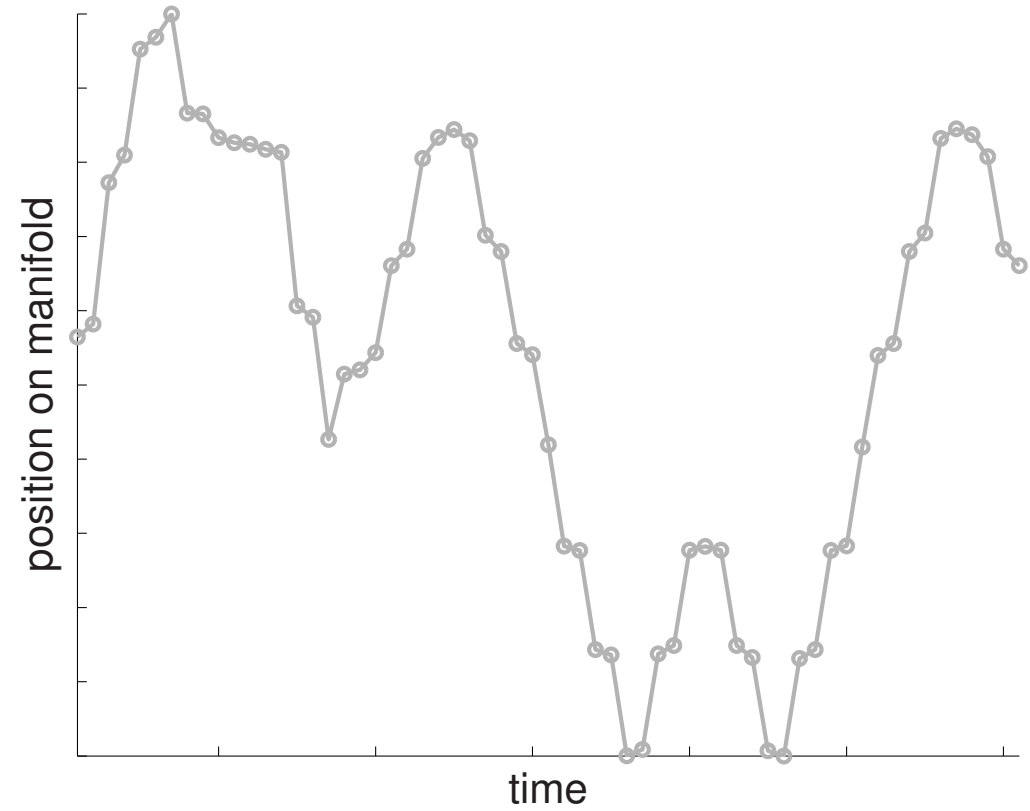
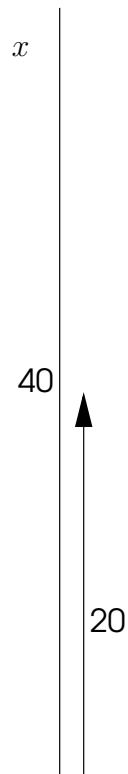
- Use learned K with kernel PCA.

Semidefinite Embedding (SDE)



- It works.
- Reduces dimensionality reduction to a constrained optimization problem.

Dimensionality Reduction Maps



Subjective Representation

- How do we extract a representation like (x, y, θ) ?
 - Low dimensional description of observations?

Dimensionality Reduction

- Respects actions as simple transformations?

Action Respecting Embedding

Overview

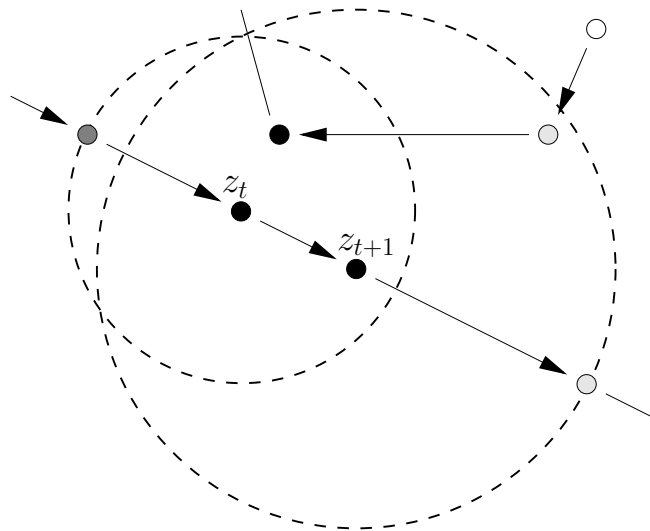
- What is a Map?
- Subjective Mapping
- Action Respecting Embedding (ARE)
 - Non-Uniform Neighborhoods
 - Action Respecting Constraints
- Results
- Future Work

Action Respecting Embedding (ARE)

- Like SDE, learns a kernel matrix K through optimization.
- Uses the kernel matrix K with kernel PCA.
- Exploits the action data (u_t) in two ways.
 - Non-uniform neighborhood graph.
 - Action respecting constraints.

Non-Uniform Neighborhoods

- Actions only have a small effect on the robot's pose.
- If $z_t \xrightarrow{u_t} z_{t+1}$, then $\Phi(z_t)$ and $\Phi(z_{t+1})$ should be close.
- Set z_t 's neighborhood size to include z_{t-1} and z_{t+1} .



- Neighbor graph uses this non-uniform neighborhood size.

Action Respecting Constraints

- Constrain the representation so that actions must be **simple transformations**.
- What's a simple transformation?
 - Linear transformations are simple.

$$f(x) = Ax + b$$

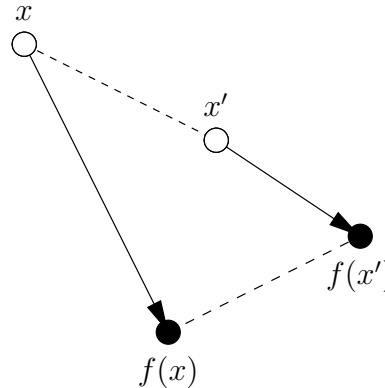
- Distance preserving ones are just slightly simpler.

$$f(x) = Ax + b \text{ where } A^T A = I.$$

i.e., rotation and translation, but no scaling.

Action Respecting Constraints

- Distance preserving $\iff ||f(x) - f(x')|| = ||x - x'||.$



- For our representation, if $u_i = u_j = u$,

$$||f_u(\Phi(z_i)) - f_u(\Phi(z_j))|| = ||\Phi(z_i) - \Phi(z_j)||$$

$$||\Phi(z_{i+1}) - \Phi(z_{j+1})|| = ||\Phi(z_i) - \Phi(z_j)||$$

$$K_{(i+1)(i+1)} - 2K_{(i+1)(j+1)} + K_{(j+1)(j+1)} = K_{ii} - 2K_{ij} + K_{jj}$$

Action Respecting Embedding

- Optimization Problem:

Maximize: $\text{Tr}(K)$

Subject to: $K \succeq 0$

$$\sum_{ij} K_{ij} = 0$$

$$\forall i, j \quad \eta_{ij} > 0 \vee [\eta^T \eta]_{ij} > 0 \Rightarrow$$

$$K_{ii} - 2K_{ij} + K_{jj} = \|z_i - z_j\|^2$$

$$\forall i, j \quad u_i = u_j \Rightarrow$$

$$K_{(i+1)(i+1)} - 2K_{(i+1)(j+1)} + K_{(j+1)(j+1)} = \\ K_{ii} - 2K_{ij} + K_{jj}$$

where η is the non-uniform neighborhood graph.

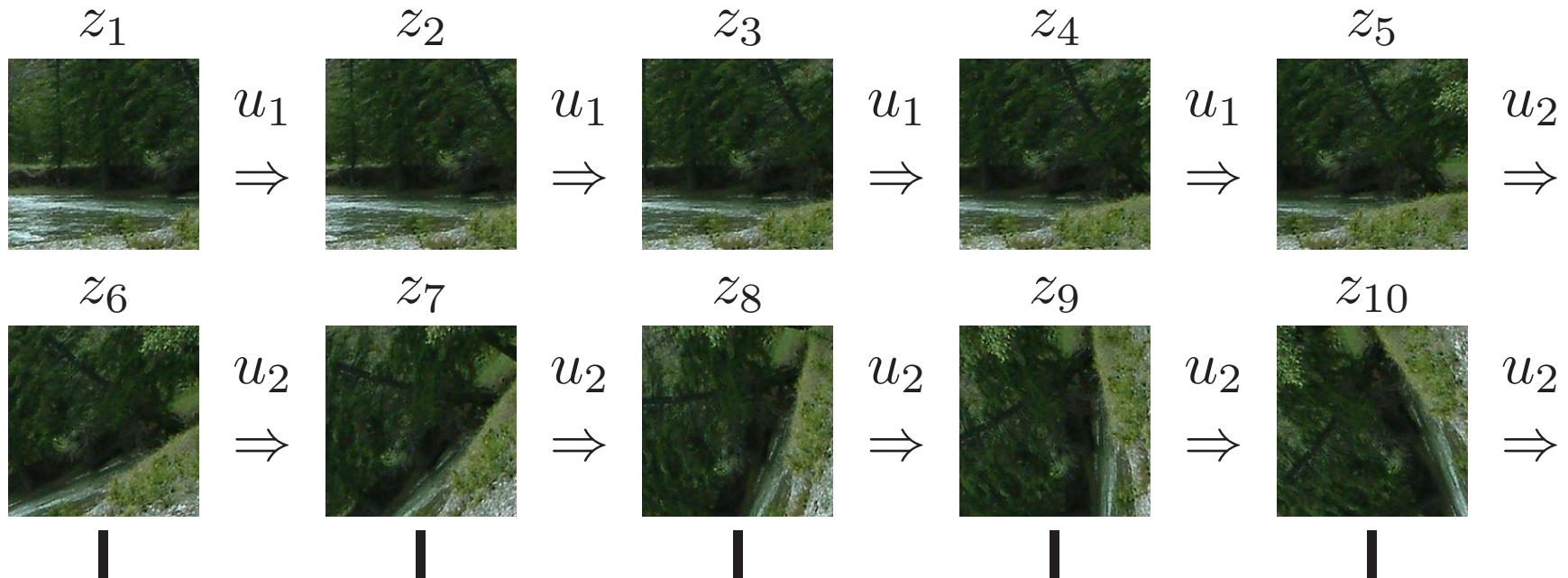
- Use learned K with kernel PCA to extract $x_1, \dots, x_T$.

Overview

- What is a Map?
- Subjective Mapping
- Action Respecting Embedding (ARE)
- Results
- Future Work

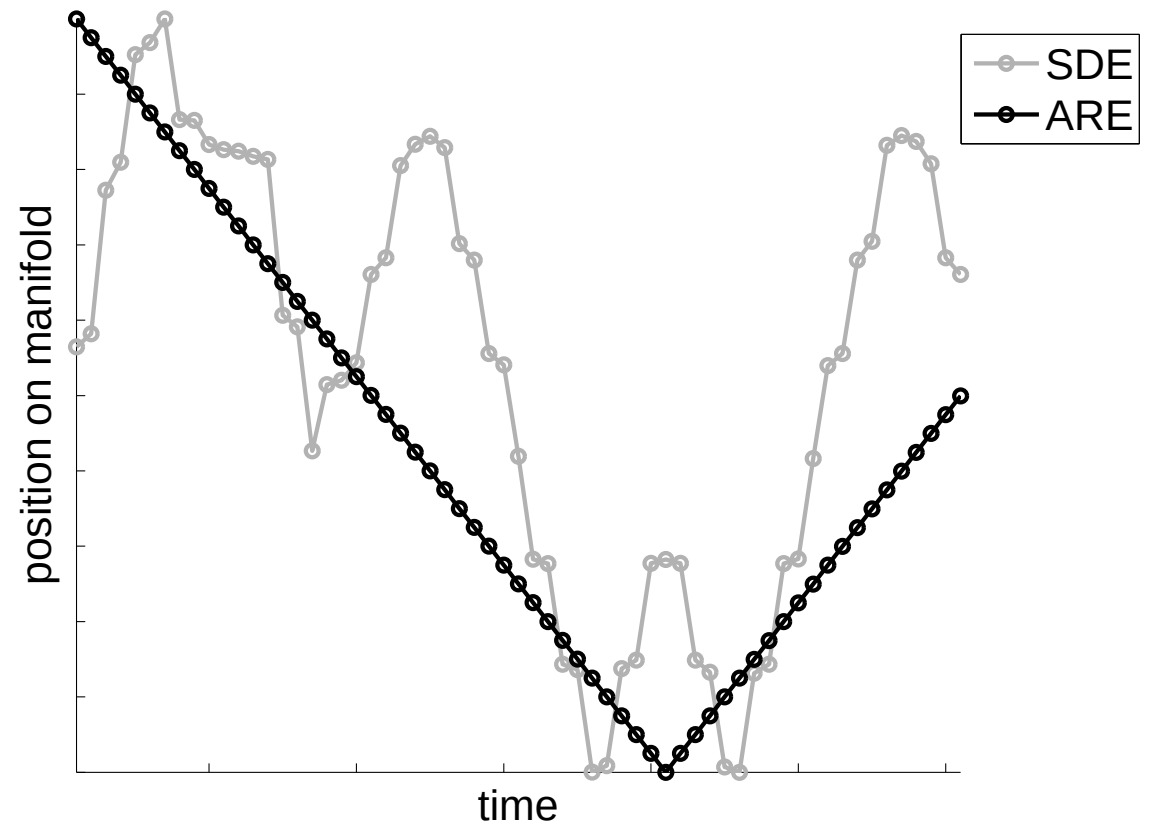
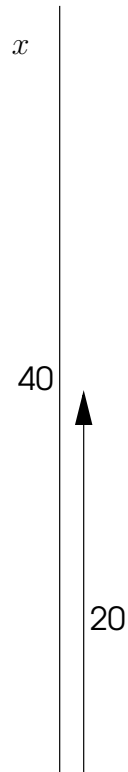
ImageBot

- Construct a map from a stream of input.

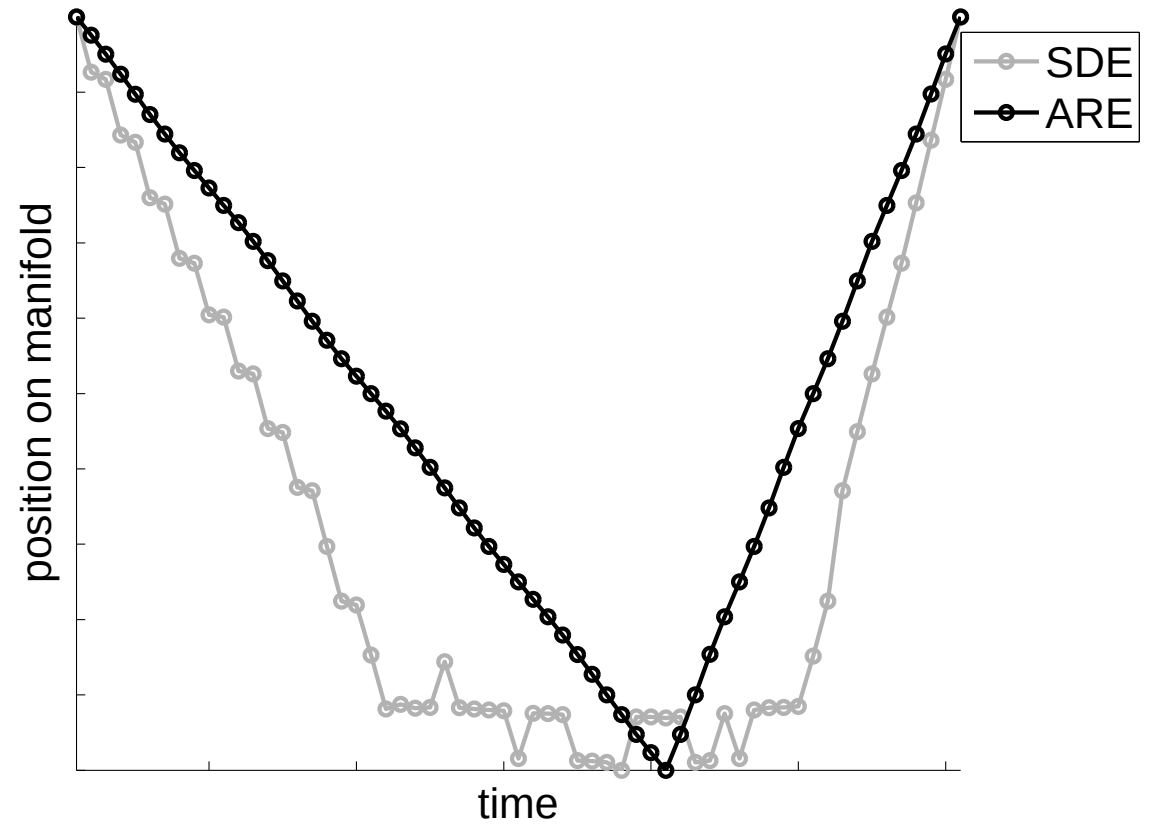
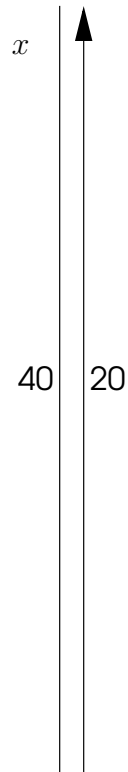


- Actions are labels with no semantics.
- No image features, just high-dimensional vectors.

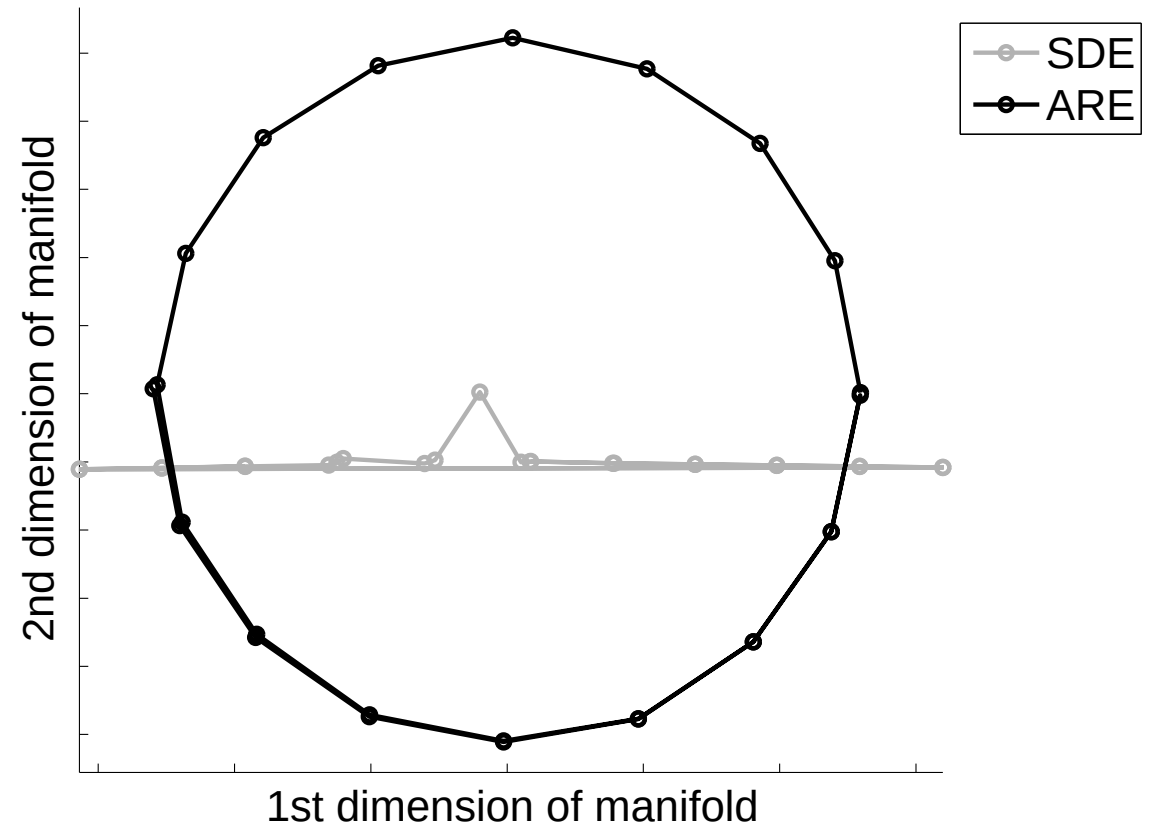
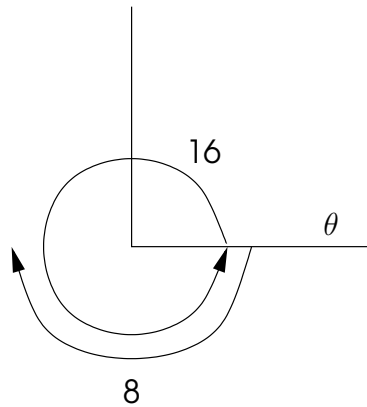
Learned Representations



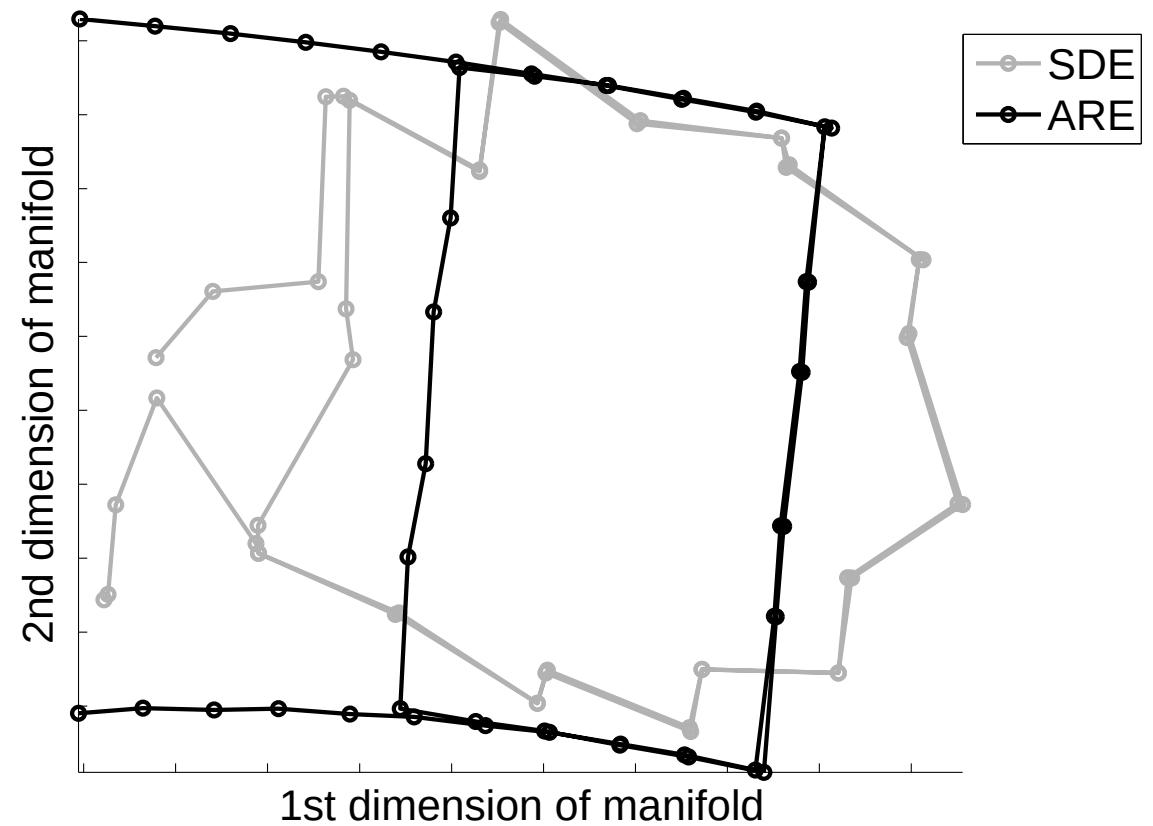
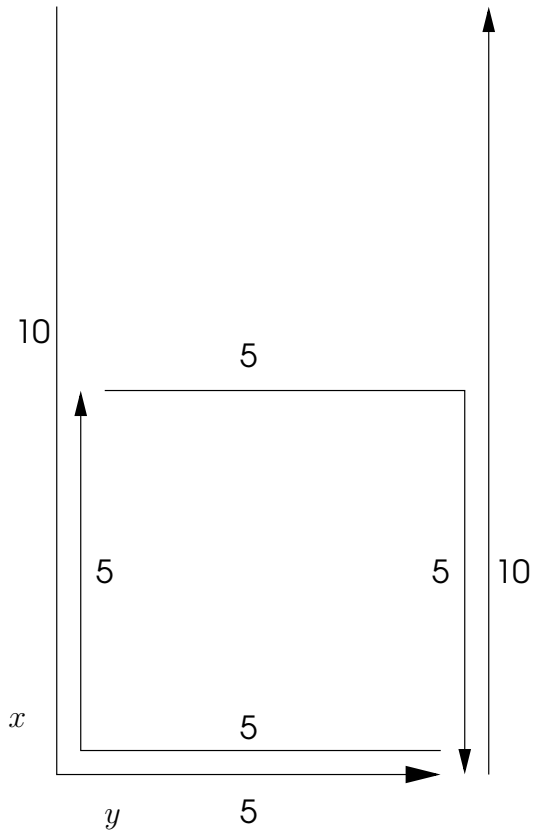
Learned Representations



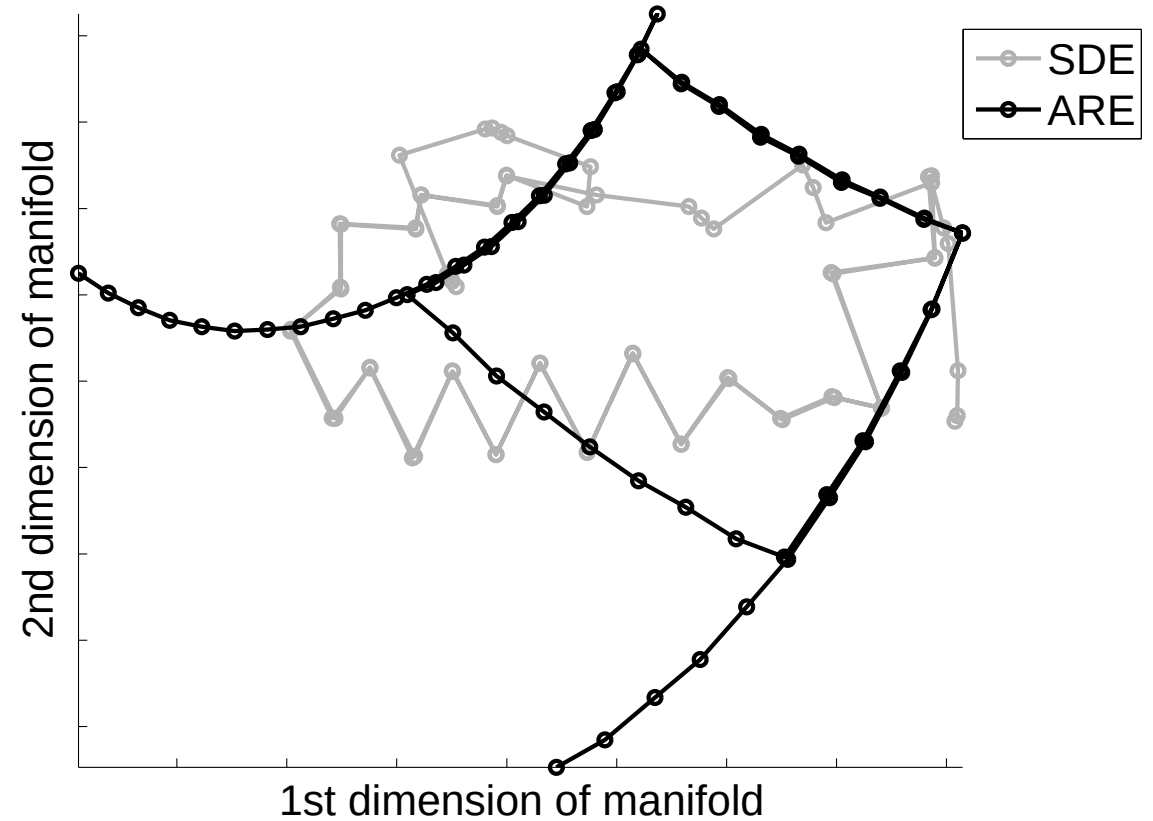
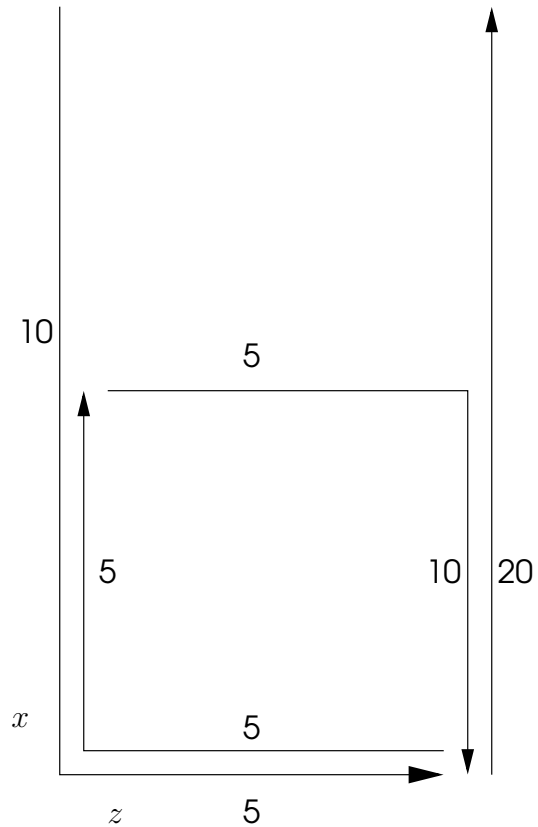
Learned Representations



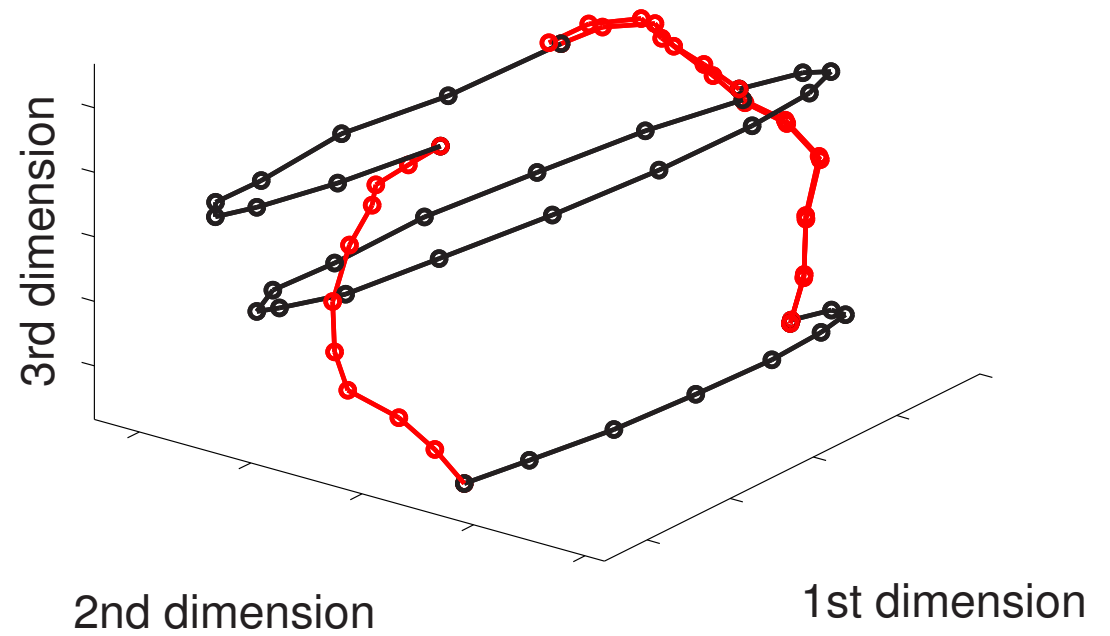
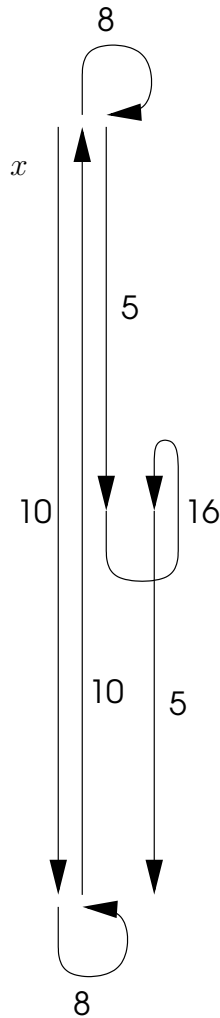
Learned Representations



Learned Representations



Learned Representations



Subjective Mapping

- Is it a map?
 - Planning
 - Extract the distance preserving operators: $f_u : X \rightarrow X$.
 - Search for an action sequence.
 - Localization
 - Extract motion and sensor models.
- $$P(x_{t+1}|x_t, u_t) \quad P(z_t|x_t)$$
- Monte Carlo localization (MCL), a.k.a. particle filtering.

Summary

- ARE automatically extracts a representation from only subjective experience of observations and actions.
- The representation is low-dimensional, action-respecting.
- It has all of the qualities of a map.
 - Can be used for [planning](#).
 - Can be used for [localization](#).

Future Work

- Faster ARE. Customized solver?
- Better ARE.
 - Learning representations with walls.
 - State aliasing.
 - Continuous actions.
- Using ARE.
 - Heuristics for planning with ARE.
 - Behavior learning with ARE representations.
- Subjective maps for real robots.

Questions?

Extra Slides

Planning

- Problem: from x_T find a sequence of actions that will reach x_* where $x_* \in \{x_1, \dots, x_T\}$.
- Need a Model: If we take action u from x_t , what is x_{t+1} ?
 - ARE guarantees the existence of a distance preserving function, f_u ,

$$\forall t \quad u_t = u \Rightarrow f_u(x_t) = x_{t+1}.$$

- We just need to extract f_u from the representation.

Planning with Procrustes



- Variation on the extended orthonormal Procrustes problem.

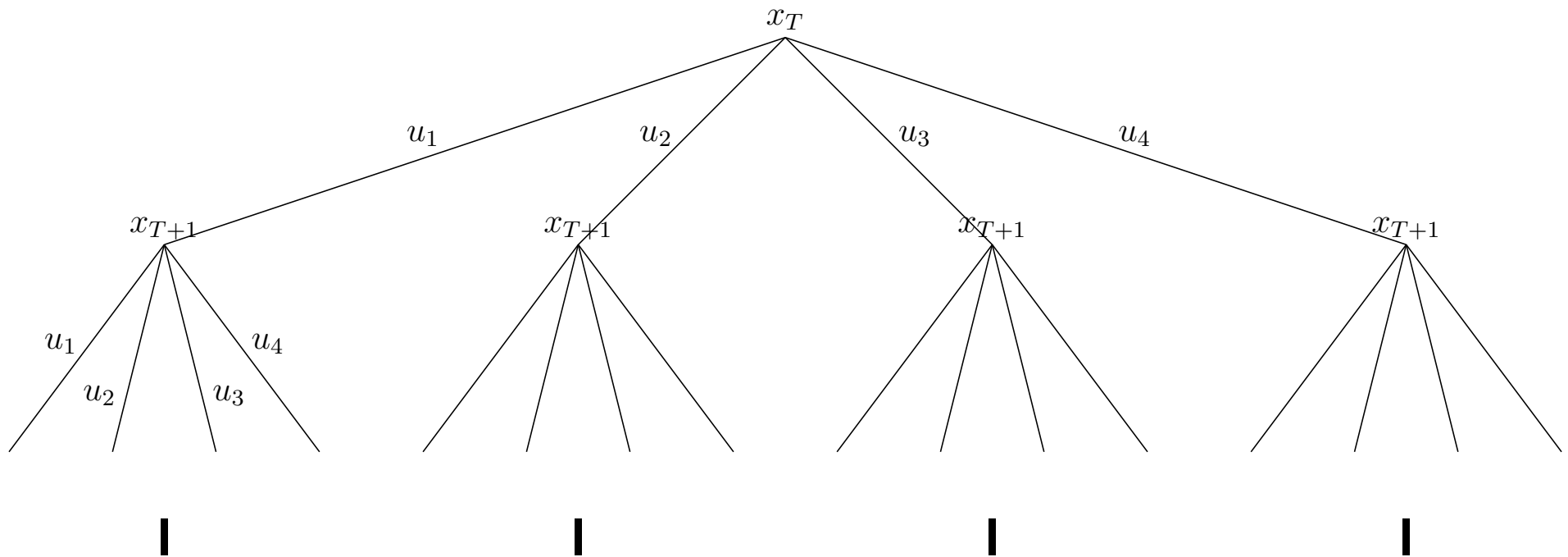
$$\begin{array}{ll} \text{Minimize:} & \sum_t \|Ax_t + b - x_{t+1}\|^2 \cdot \mathbf{1}_{u_t=u} \\ & A, b \end{array}$$

$$\text{Subject to:} \quad A^T A = I$$

- Although not convex, it has an analytic solution!

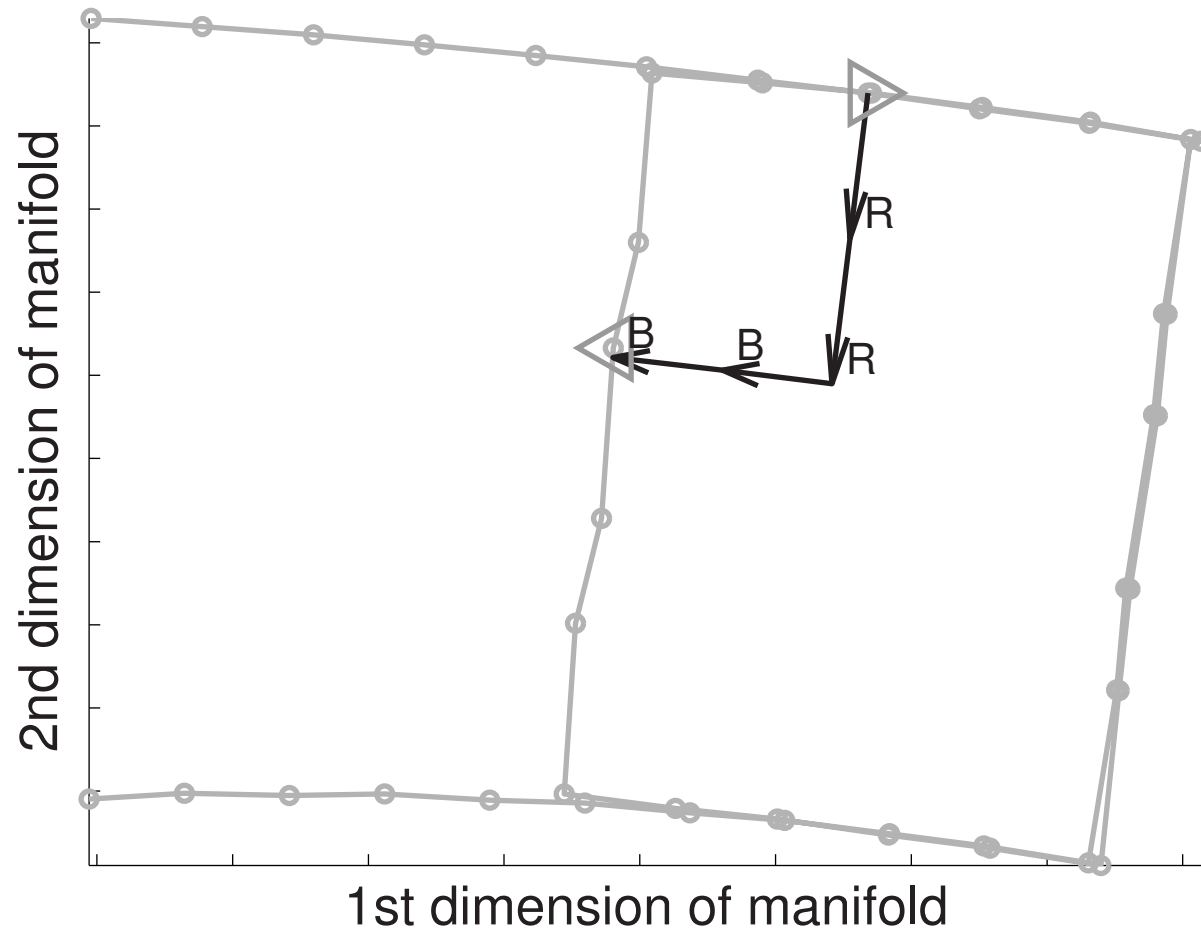
Planning with Search

- Using our action models, we can search for a plan.

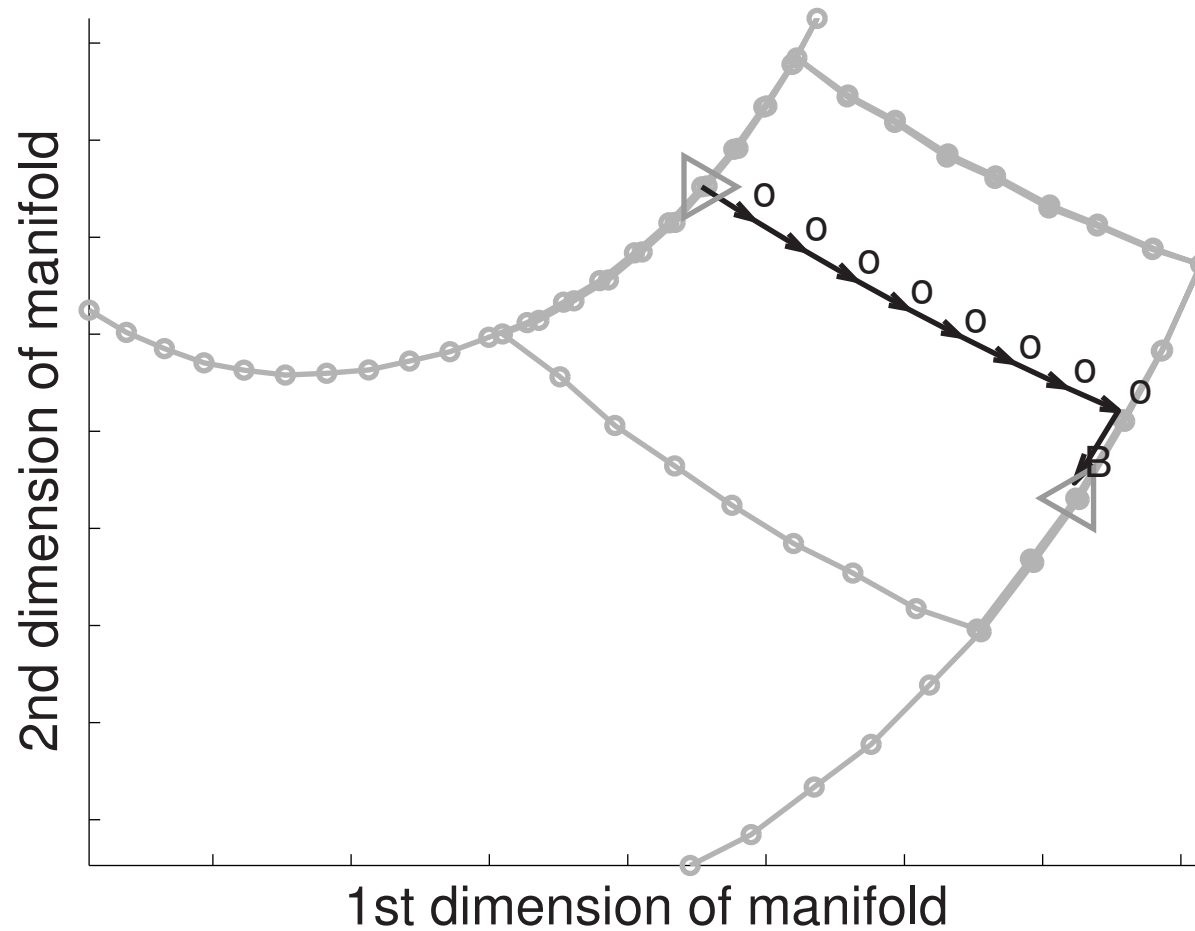


- Simple iterative deepening depth-first search.

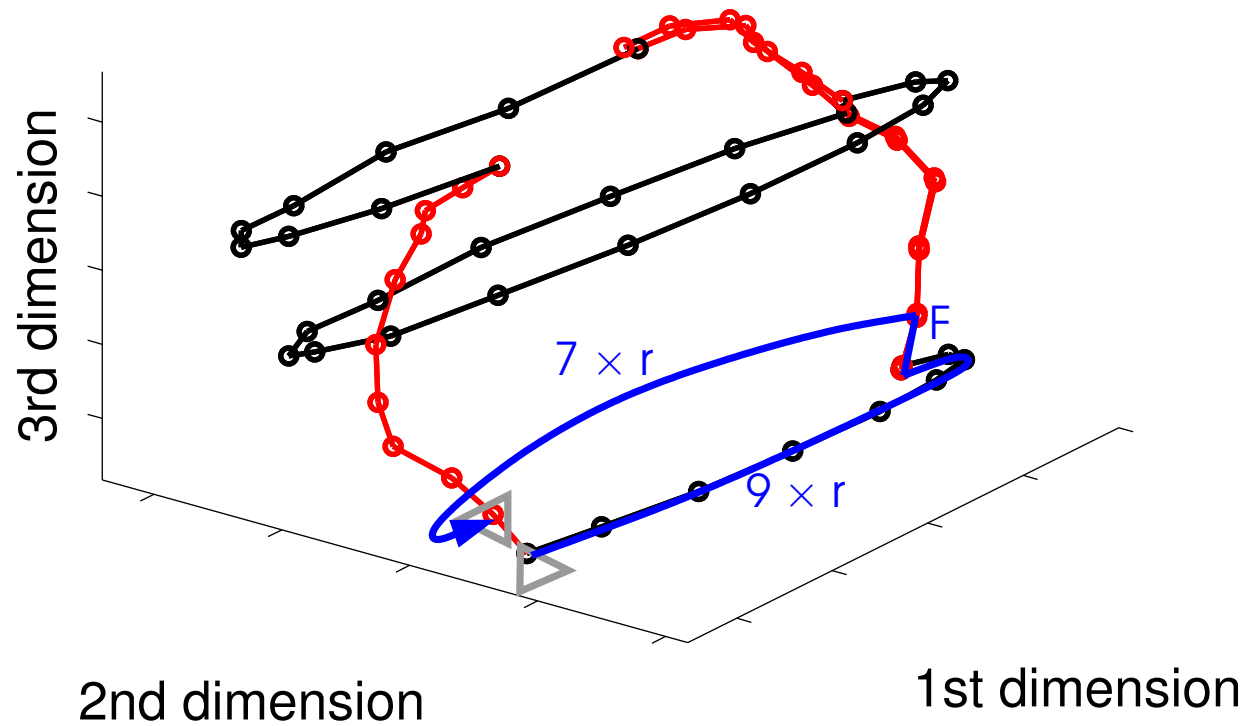
Planning Results



Planning Results



Planning Results



Return

Localization

- Maps are Models.
 - Motion: $P(x_{t+1}|x_t, u_t)$.
 - Sensor: $P(z_t|x_t)$.
- Extract noisy action models from experience.
 - Noise is encoded in the later principal components.
 - Procrustes extracts the mean.
 - Regression errors define the variance.
- Extract sensor model based on correlation between image distances and distance in the representation.
- Monte Carlo localization (MCL).

Localization Results

- Noisy actions in IMAGEBOT.
- No ground truth with subjective representations.
- Choose the “closest” point in the training trajectory.

Trajectory	Mean Error (Pixels)		
	Oracle	ARE	Random
Straight line	4.82	10.39	86.83
“A” with translation	3.62	14.81	104.56
“A” with zoom	1.71	19.58	84.67

Return